

BSc Løsning om Hesel simulering og PDE

$$V_a(t, r) = \frac{1}{n_a} \int v f_a(t, r, v) dv = \langle v \rangle_a : \text{Gennemsnits hastighed for partikel } v$$

$m \cdot \langle v \rangle_a$ er relateret til temperatur

$$n \cdot \langle v \rangle_a = \frac{3}{2} T \text{ for koordinat system med } v=0$$

$$T_a(t, r) = \frac{1}{n_a} \int \frac{m_a}{2} (v - V_a)^2 f_a(t, r, v) dv = \frac{m_a}{2} \langle (v - V_a)^2 \rangle$$

$$\star \frac{\partial n_a}{\partial t} + \text{div}(n_a V) = 0 : \text{Transportligningen for plasma}$$

1D Eksempler

1. Normalfordeling med konstant V felt. Vi finder V

$$n_a(x, t) = \mathcal{N}(x=x, \mu=t, \sigma=\frac{\sqrt{2}}{2}) = \frac{1}{\sqrt{\pi}} e^{-(x-t)^2}$$

$$\frac{d}{dt} n_a(x, t) = \frac{1}{\sqrt{\pi}} \cdot (2(x-t)) e^{-(x-t)^2}, \quad p = x-t, \quad g(p) = \frac{1}{\sqrt{\pi}} e^{-p^2}$$

$$\Rightarrow \frac{\partial n_a}{\partial t} = g'(p) \cdot \partial_t(p) = g'(p)(-1) = -g'(p)$$

$$\Rightarrow \frac{\partial n_a}{\partial x} = g'(p) \cdot \partial_x(p) = g'(p)$$

$$-\partial_x n_a + \partial_x(V n_a) = 0 \Rightarrow \partial_x((V-1)n_a) = 0 \Leftrightarrow V(x, t) = 1 + \frac{c(t)}{n_a(x, t)}$$

$$c(t) = 0 \Leftrightarrow V(x, t) = 1 + t \cdot \mathbb{R}$$

2. Oscillerende normalfordelt mængde

$$V(x, t) = \cos(t)$$

$$\frac{\partial n}{\partial t} + V(x, t) \frac{\partial n}{\partial x} = 0 \Leftrightarrow x(t) = \int_0^t V(s) ds = \sin(t) + c \quad "c=0" \Rightarrow x(t) = \sin(t)$$

$$n(x, x(t)) = \frac{1}{\sqrt{\pi}} e^{-(x - \sin(t))^2}$$